

# Iterated Thue–Morse Prefix Sums and the Fabius Function

Exact centering, a sharp  $n2^{-n}$  law, and faster recentered approximations

A complementary report to the Fabius–Rvachev synthesis

Based on Vladimir Reshetnikov’s `ProveIt` formal development

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## Abstract

Let  $\varepsilon_m = (-1)^{\text{wt}_2(m)}$  be the signed Thue–Morse sequence and let  $\sigma_{k+1}(m) = \sum_{j=0}^m \sigma_k(j)$  be its iterated *inclusive* prefix sums. The formal development in the `ProveIt` repository proves that

$$\mathcal{A}_n(x) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}, \quad 0 \leq x \leq 1,$$

converges pointwise to the bounded Fabius function  $F(x)$ . The existing exposition deliberately asserts no rate. This report supplies one.

The coefficient carrying  $\mathcal{A}_n(x)$  is the value of a polynomial histogram at a cell center, but that center is not  $x - 1$ . Its exact displacement is

$$\delta_n(x) = \frac{n + 2 - 2\{2^n x\}}{2^{n+1}} = \frac{n}{2^{n+1}} + \frac{1 - \{2^n x\}}{2^n}.$$

At the histogram centers themselves, a Fourier-tail factorization and a cardinal moment calculation give the uniform estimate

$$|\mathcal{W}_n(\xi_{n,m}) - \text{up}(\xi_{n,m})| \leq \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}.$$

Consequently, uniformly for  $x \in [0, 1]$  and  $n \geq 3$ ,

$$\mathcal{A}_n(x) = F(x) + F'(x)\delta_n(x) + \frac{1}{2}F''(x)\delta_n(x)^2 + \rho_n(x),$$

where

$$|\rho_n(x)| \leq \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n} + \frac{4}{3}(n+2)^3 8^{-n}.$$

In fact the entire normalized error profile converges uniformly:

$$\frac{2^n}{n}(\mathcal{A}_n - F) \longrightarrow \frac{1}{2}F' \quad \text{in } L^\infty[0, 1].$$

Consequently the uniform error has the sharp constant

$$\boxed{\lim_{n \rightarrow \infty} \frac{2^n}{n} \|\mathcal{A}_n - F\|_{L^\infty[0,1]} = 1.}$$

The factor  $n$  is therefore not an intrinsic weakness of the Thue–Morse coefficients. It is a deterministic centering drift. The histogram itself has uniform error asymptotic to  $2^{-n}$ , while linear interpolation through the correct coefficient centers has the explicit faster bound  $(4n/3 + 25/9)4^{-n}$ . The report also records exact endpoint and midpoint formulas, a three-scale asymptotic expansion containing the binary phase  $\{2^n x\}$ , and a roadmap for formalizing the new quantitative theorem in Lean.

## Contents

|           |                                                                 |           |
|-----------|-----------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Scope, conventions, and the quantitative question</b>        | <b>3</b>  |
| <b>2</b>  | <b>The discrete hierarchy</b>                                   | <b>3</b>  |
| 2.1       | Signed Thue–Morse and inclusive summation . . . . .             | 3         |
| 2.2       | Dyadic annihilation . . . . .                                   | 4         |
| <b>3</b>  | <b>The exact polynomial bridge</b>                              | <b>4</b>  |
| 3.1       | Approximation polynomials . . . . .                             | 4         |
| 3.2       | The centered probability measure and histogram . . . . .        | 5         |
| <b>4</b>  | <b>What the floor index actually samples</b>                    | <b>6</b>  |
| <b>5</b>  | <b>Fourier factorization of the omitted tail</b>                | <b>6</b>  |
| 5.1       | The cosine products . . . . .                                   | 6         |
| 5.2       | A cardinal probability distribution . . . . .                   | 7         |
| <b>6</b>  | <b>Discrete curvature of the coefficient rows</b>               | <b>8</b>  |
| <b>7</b>  | <b>The intrinsic error at coefficient centers</b>               | <b>9</b>  |
| <b>8</b>  | <b>The sharp expansion for the Fabius sample</b>                | <b>9</b>  |
| <b>9</b>  | <b>Pointwise and uniform sharpness</b>                          | <b>11</b> |
| 9.1       | Uniform error profile and pointwise law . . . . .               | 11        |
| 9.2       | The exact uniform constant . . . . .                            | 11        |
| <b>10</b> | <b>Exact special values</b>                                     | <b>12</b> |
| 10.1      | The endpoints . . . . .                                         | 12        |
| 10.2      | An exact midpoint coefficient . . . . .                         | 12        |
| <b>11</b> | <b>The corresponding result for Rvachev’s bump</b>              | <b>13</b> |
| <b>12</b> | <b>Rates for the histogram and for recentered interpolation</b> | <b>14</b> |
| 12.1      | The step histogram . . . . .                                    | 14        |
| 12.2      | Linear interpolation through the true centers . . . . .         | 14        |
| <b>13</b> | <b>Why the constants take these values</b>                      | <b>15</b> |
| <b>14</b> | <b>A Lean formalization roadmap</b>                             | <b>15</b> |
| <b>15</b> | <b>Conclusion</b>                                               | <b>16</b> |
| <b>A</b>  | <b>Notation crosswalk</b>                                       | <b>17</b> |
| <b>B</b>  | <b>A compact statement of the main estimates</b>                | <b>17</b> |

# 1 Scope, conventions, and the quantitative question

The purpose of this report is narrow but structurally revealing. The existing Fabius–Rvachev synthesis develops the discrete Thue–Morse construction in full: the prefix hierarchy, its binomial convolution kernel, the dyadic zero runs, the formal generating series, the corrected normalization, the exact identification with the polynomial histogram, and the resulting pointwise convergence [5, 6]. The corresponding Lean module `ThueMorseApproximation.lean` explicitly says that no quantitative uniform rate is asserted. We keep every convention of that development and add the missing estimate.

Throughout,  $F$  is the bounded Fabius distribution function, equal to 0 on  $(-\infty, 0]$  and to 1 on  $[1, \infty)$ , and  $\text{up}$  is Rvachev’s even compactly supported bump,

$$\text{up}(y) = F(1 - |y|), \quad F(x) = \text{up}(x - 1) \quad (0 \leq x \leq 1). \quad (1)$$

We use the Fourier transform

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad \text{sinc } z = \frac{\sin z}{z}. \quad (2)$$

The derivative bounds proved sharply in the formal development are

$$\|F^{(r)}\|_{\infty} \leq 2^{\binom{r+1}{2}}, \quad \|\text{up}^{(r)}\|_{\infty} \leq 2^{\binom{r+1}{2}}. \quad (3)$$

We shall use, in particular,

$$0 \leq F' \leq 2, \quad \|F''\|_{\infty} \leq 8, \quad \|F'''\|_{\infty} \leq 64, \quad (4)$$

and  $F'(1/2) = 2$ .

The central question is not merely whether the prefix sequence converges, but which of three mechanisms controls its error:

1. the difference between the finite coefficient histogram and  $\text{up}$  at the exact coefficient centers;
2. the discretization incurred by replacing a smooth function with a step histogram;
3. the displacement between the center selected by the published floor index and the intended evaluation point.

The answer is unexpectedly clean. These effects have orders  $n4^{-n}$ ,  $2^{-n}$ , and  $n2^{-n}$ , respectively. The third dominates the particular floor-sampled approximation to  $F$ .

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**Status of formal verification.** The exact prefix–coefficient identity and the qualitative convergence used here are already formalized in Lean. The quantitative center estimate, the asymptotic expansion, and the sharp constant 1 are new deductions in this report. They are presented with complete conventional proofs, but they have not been added to the repository’s Lean files as of the snapshot date.

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## 2 The discrete hierarchy

### 2.1 Signed Thue–Morse and inclusive summation

Let

$$\varepsilon_m = (-1)^{\text{wt}_2(m)}, \quad (5)$$

where  $\text{wt}_2(m)$  is the number of 1’s in the binary expansion of  $m$ . Define

$$\sigma_0(m) = \varepsilon_m, \quad \sigma_{k+1}(m) = \sum_{j=0}^m \sigma_k(j). \quad (6)$$

The summation is inclusive. Thus

$$\sigma_{k+1}(m+1) - \sigma_{k+1}(m) = \sigma_k(m+1), \quad (7)$$

with the lower-order value evaluated at the new index. This one-place shift is responsible for the order correction that appears later.

Repeated summation gives the discrete B-spline convolution law

$$\sigma_k(m) = \sum_{r=0}^m \binom{m-r+k-1}{k-1} \varepsilon_r, \quad k \geq 1. \quad (8)$$

Equivalently,

$$\sigma_{k+1}(m) = \sum_{r=0}^m \binom{m-r+k}{k} \varepsilon_r. \quad (9)$$

The kernel is the coefficient sequence of  $(1-z)^{-k}$ .

The formal generating functions make the same statement in one line. On the first dyadic block,

$$K_r(z) = \sum_{m=0}^{2^r-1} \varepsilon_m z^m = \prod_{j=0}^{r-1} (1 - z^{2^j}), \quad (10)$$

and coefficientwise the infinite signed series is

$$K(z) = \prod_{j \geq 0} (1 - z^{2^j}). \quad (11)$$

Hence the  $k$ -fold prefix series is formally

$$S_k(z) = \sum_{m \geq 0} \sigma_k(m) z^m = \frac{K(z)}{(1-z)^k}. \quad (12)$$

No analytic convergence is involved in this identity.

## 2.2 Dyadic annihilation

The product in (10) has a zero of order  $r$  at  $z = 1$ . Equivalently, a complete block of length  $2^r$  annihilates every polynomial in  $m$  of degree below  $r$ . Applied to (8), this gives the exact zero run

$$\sigma_k(m) = 0 \quad \text{for} \quad 2^r - k \leq m \leq 2^r - 1, \quad 1 \leq k \leq r, \quad (13)$$

with sharp boundary values

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \quad \sigma_k(2^r) = -1. \quad (14)$$

These identities explain both the long flat regions of the iterated prefixes and the exceptional right endpoint of the unit-interval grid.

## 3 The exact polynomial bridge

### 3.1 Approximation polynomials

Put

$$P_n(z) = \prod_{j=1}^n (1 + z + \cdots + z^{2^j-1}) = \sum_{m=0}^{g_n} a_{n,m} z^m, \quad (15)$$

with  $P_0 = 1$ . Its degree and total mass are

$$g_n = \sum_{j=1}^n (2^j - 1) = 2^{n+1} - n - 2, \quad P_n(1) = 2^{\binom{n+1}{2}}. \quad (16)$$

The coefficients count restricted sums

$$a_{n,m} = \# \left\{ (s_1, \dots, s_n) : 0 \leq s_j < 2^j, \sum_{j=1}^n s_j = m \right\}. \quad (17)$$

They are nonnegative and symmetric:  $a_{n,m} = a_{n,g_n-m}$ .

The denominator-cleared product identity is

$$(1-z)^{n+1} P_n(z) = \prod_{j=0}^n (1-z^{2^j}). \quad (18)$$

Comparing (18) with (12) gives the exact coefficient bridge

$$\boxed{\sigma_{n+1}(m) = a_{n,m} \quad \text{whenever} \quad 0 \leq m < 2^{n+1}.} \quad (19)$$

The restriction is real: at  $m = 2^{n+1}$  the left side is  $-1$  by (14), while the polynomial coefficient is zero.

### 3.2 The centered probability measure and histogram

Let

$$h_n = 2^{-n}, \quad \xi_{n,m} = \frac{2m - g_n}{2^{n+1}} = h_n \left( m - \frac{g_n}{2} \right). \quad (20)$$

The centers form an arithmetic progression of mesh  $h_n$ . Define the atomic probability measure

$$\mu_n = \sum_{m \in \mathbb{Z}} p_{n,m} \delta_{\xi_{n,m}}, \quad p_{n,m} = \frac{a_{n,m}}{2^{\binom{n+1}{2}}}, \quad (21)$$

where  $a_{n,m} = 0$  outside  $0 \leq m \leq g_n$ . Let

$$\beta_h(x) = \frac{1}{h} \mathbf{1}_{[-h/2, h/2]}(x) \quad (22)$$

with the harmless half-weight convention at the two endpoints. The polynomial histogram is

$$\mathcal{W}_n = \mu_n * \beta_{h_n}. \quad (23)$$

At a center only its own atom contributes, and therefore

$$\mathcal{W}_n(\xi_{n,m}) = \frac{p_{n,m}}{h_n} = \frac{a_{n,m}}{2^{\binom{n}{2}}}. \quad (24)$$

Combining (19) and (24) yields the formalized exact identification

$$\frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}} = \mathcal{W}_n(\xi_{n,m}), \quad 0 \leq m < 2^{n+1}. \quad (25)$$

## 4 What the floor index actually samples

The corrected prefix sample for the bump is

$$\mathcal{J}_n(t) = \frac{\sigma_{n+1}(\lfloor 2^{n+1}t \rfloor)}{2^{\binom{n}{2}}}, \quad 0 \leq t \leq 1. \quad (26)$$

For  $t < 1$ , (25) gives

$$\mathcal{J}_n(t) = \mathcal{W}_n(\xi_{n, \lfloor 2^{n+1}t \rfloor}). \quad (27)$$

The Fabius version is the first-half rescaling

$$\mathcal{A}_n(x) = \mathcal{J}_n(x/2) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}, \quad 0 \leq x \leq 1. \quad (28)$$

**Lemma 4.1** (Exact center drift). *For  $x \in [0, 1]$ , put*

$$\theta_n(x) = \{2^n x\}, \quad \delta_n(x) = \frac{n+2-2\theta_n(x)}{2^{n+1}}. \quad (29)$$

Then

$$\xi_{n, \lfloor 2^n x \rfloor} = x - 1 + \delta_n(x), \quad (30)$$

and

$$\delta_n(x) = \frac{n}{2^{n+1}} + \frac{1 - \theta_n(x)}{2^n}, \quad \frac{n}{2^{n+1}} < \delta_n(x) \leq \frac{n+2}{2^{n+1}}. \quad (31)$$

*Proof.* Use  $g_n = 2^{n+1} - n - 2$  and  $\lfloor 2^n x \rfloor = 2^n x - \theta_n(x)$  in (20):

$$\begin{aligned} \xi_{n, \lfloor 2^n x \rfloor} &= \frac{2(2^n x - \theta_n(x)) - (2^{n+1} - n - 2)}{2^{n+1}} \\ &= x - 1 + \frac{n+2-2\theta_n(x)}{2^{n+1}}. \end{aligned}$$

The decomposition and bounds follow from  $0 \leq \theta_n(x) < 1$ . □

The dominant term in (31) is  $n2^{-n-1}$ . It comes from the deficit  $n+2$  in the degree  $g_n = 2^{n+1} - n - 2$ , not from random rounding. The binary fractional part contributes the lower-order oscillation  $(1 - \theta_n(x))2^{-n}$ .

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**The source of the rate.** The approximation does not read the histogram near  $x - 1$  and then incur a single-cell error. It reads a center roughly  $n/2$  cells to the right. Since  $F'$  is generally nonzero, a smooth function changes by order  $(n/2)h_n = n2^{-n-1}$  across that displacement. Any proof that ignores the exact center formula can at best obscure this leading term.

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## 5 Fourier factorization of the omitted tail

### 5.1 The cosine products

The centered uniform variable on the  $2^j$  lattice points of spacing  $h_n$  has Fourier transform

$$\frac{\sin(\pi 2^j h_n \xi)}{2^j \sin(\pi h_n \xi)} = \prod_{r=0}^{j-1} \cos(\pi 2^r h_n \xi). \quad (32)$$

Multiplying over  $j = 1, \dots, n$  and collecting equal cosine factors gives

$$\widehat{\mu}_n(\xi) = \prod_{\ell=1}^n \cos\left(\frac{\pi\xi}{2^\ell}\right)^\ell. \quad (33)$$

The Rvachev bump has the infinite product

$$\widehat{\text{up}}(\xi) = \prod_{\ell=1}^{\infty} \cos\left(\frac{\pi\xi}{2^\ell}\right)^\ell. \quad (34)$$

This is equivalent to the more familiar infinite sinc product.

Set  $h = h_n$ . Reindexing the tail of (34) gives

$$\begin{aligned} \prod_{\ell=n+1}^{\infty} \cos\left(\frac{\pi\xi}{2^\ell}\right)^\ell &= \prod_{r=1}^{\infty} \cos\left(\frac{\pi h\xi}{2^r}\right)^{n+r} \\ &= \widehat{\text{up}}(h\xi) \left( \prod_{r=1}^{\infty} \cos\left(\frac{\pi h\xi}{2^r}\right) \right)^n \\ &= \widehat{\text{up}}(h\xi) \text{sinc}(\pi h\xi)^n. \end{aligned} \quad (35)$$

In the last line we used  $\prod_{r \geq 1} \cos(z/2^r) = \sin z/z$ .

Let  $D_h \text{up}(x) = h^{-1} \text{up}(x/h)$  and define the tail density

$$q_n = \beta_h^{*n} * D_h \text{up}. \quad (36)$$

The Fourier identity (35) is precisely

$$\boxed{\text{up} = \mu_n * q_n}. \quad (37)$$

Thus the limit is obtained from the discrete polynomial law by adding  $n$  independent uniform jitters of width  $h$  and an independent scaled copy of the whole Rvachev law.

## 5.2 A cardinal probability distribution

At the mesh points define

$$\lambda_{n,k} = h q_n(kh), \quad k \in \mathbb{Z}. \quad (38)$$

The  $\lambda_{n,k}$  form a symmetric probability distribution on  $\mathbb{Z}$ . Indeed, write  $q_n = \beta_h * \rho_n$ . Since translates of the box tile the line,

$$h \sum_{k \in \mathbb{Z}} \beta_h(kh - s) = 1 \quad (39)$$

for almost every  $s$ . Integration against  $\rho_n$  yields

$$\sum_{k \in \mathbb{Z}} \lambda_{n,k} = 1, \quad \lambda_{n,-k} = \lambda_{n,k}. \quad (40)$$

Equation (37), together with  $p_{n,m} = h \mathcal{W}_n(\xi_{n,m})$ , now becomes the exact cardinal averaging identity

$$\text{up}(\xi_{n,m}) = \sum_{k \in \mathbb{Z}} \lambda_{n,k} \mathcal{W}_n(\xi_{n,m-k}). \quad (41)$$

The value of the smooth limit at a center is therefore a symmetric local average of the neighboring coefficient heights.

The second moment of this discrete averaging law is also exact.

**Lemma 5.1** (Cardinal second moment). *For  $n \geq 3$ ,*

$$\sum_{k \in \mathbb{Z}} k^2 \lambda_{n,k} = \frac{n}{12} + \frac{1}{9}. \quad (42)$$

*Proof.* After scaling by  $h$ , let

$$\kappa_n = \beta_1^{*n} * \text{up}, \quad \lambda_{n,k} = \kappa_n(k).$$

Its Fourier transform is  $\widehat{\kappa}_n(\xi) = \text{sinc}(\pi\xi)^n \widehat{\text{up}}(\xi)$ . Poisson summation, differentiated twice, gives

$$-4\pi^2 \sum_{k \in \mathbb{Z}} k^2 \kappa_n(k) = \sum_{j \in \mathbb{Z}} \widehat{\kappa}_n''(j).$$

At every nonzero integer  $j$ , the factor  $\text{sinc}(\pi\xi)^n$  has a zero of order  $n \geq 3$ , so the second derivative vanishes there. Only  $j = 0$  remains, and hence the discrete second moment equals the ordinary second moment of  $\kappa_n$ .

The variance of a uniform variable on  $[-1/2, 1/2]$  is  $1/12$ . The cosine product (34) represents the Rvachev law as the sum of  $\ell$  independent signs of amplitude  $2^{-\ell-1}$  at every level  $\ell$ ; therefore

$$\text{Var}(\text{up}) = \sum_{\ell=1}^{\infty} \frac{\ell}{2^{2\ell+2}} = \frac{1}{4} \sum_{\ell=1}^{\infty} \ell 4^{-\ell} = \frac{1}{9}.$$

Variances add under convolution, giving (42).  $\square$

## 6 Discrete curvature of the coefficient rows

Write

$$w_{n,m} = \frac{a_{n,m}}{2^{\binom{n}{2}}}, \quad (43)$$

with  $w_{n,m} = 0$  outside the polynomial support. Thus  $w_{n,m} = \mathcal{W}_n(\xi_{n,m})$ .

We first need a uniform coefficient bound. The recurrence

$$P_r(z) = P_{r-1}(z)(1 + z + \cdots + z^{2^r-1}) \quad (44)$$

shows that every coefficient of  $P_r$  is a partial sum of coefficients of  $P_{r-1}$ . Consequently

$$0 \leq a_{r,m} \leq P_{r-1}(1) = 2^{\binom{r}{2}}. \quad (45)$$

Equality holds on the central plateau  $g_{r-1} \leq m \leq 2^r - 1$ .

The relevant smoothness is measured by the second finite difference.

**Lemma 6.1** (Uniform discrete curvature). *For  $n \geq 3$  and every  $m \in \mathbb{Z}$ ,*

$$|w_{n,m} - 2w_{n,m-1} + w_{n,m-2}| \leq 32 \cdot 4^{-n}. \quad (46)$$

*Proof.* Factor off the last two geometric polynomials:

$$(1 - z)^2 P_n(z) = (1 - z^{2^{n-1}})(1 - z^{2^n}) P_{n-2}(z). \quad (47)$$

The coefficient on the left is  $a_{n,m} - 2a_{n,m-1} + a_{n,m-2}$ . The coefficient on the right is a signed sum of four coefficients of  $P_{n-2}$ . By (45), each has absolute value at most  $2^{\binom{n-2}{2}}$ . Division by  $2^{\binom{n}{2}}$  gives

$$4 \cdot 2^{\binom{n-2}{2} - \binom{n}{2}} = 4 \cdot 2^{-(2n-3)} = 32 \cdot 4^{-n}.$$

$\square$



**Lemma 6.2** (Symmetric discrete Taylor bound). *If a doubly infinite sequence  $(v_m)$  satisfies  $|v_m - 2v_{m-1} + v_{m-2}| \leq M$  for all  $m$ , then*

$$|v_{m+k} + v_{m-k} - 2v_m| \leq Mk^2 \quad (k \geq 0). \quad (48)$$

*Proof.* Let  $d_m = v_m - v_{m-1}$ . Consecutive first differences satisfy  $|d_m - d_{m-1}| \leq M$ . Expanding the two increments from  $m$  to  $m+k$  and from  $m-k$  to  $m$ , and pairing their first differences, yields a sum of  $1 + 3 + \dots + (2k-1) = k^2$  second differences.  $\square$

## 7 The intrinsic error at coefficient centers

We can now quantify the part of the approximation that the qualitative moving argument proof leaves hidden.

**Theorem 7.1** (Uniform center estimate). *For every  $n \geq 3$  and every  $m \in \mathbb{Z}$ ,*

$$\boxed{|\mathcal{W}_n(\xi_{n,m}) - \text{up}(\xi_{n,m})| \leq \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}.} \quad (49)$$

*Proof.* Use (41), symmetry of  $\lambda_{n,k}$ , and  $\sum_k \lambda_{n,k} = 1$ :

$$\text{up}(\xi_{n,m}) - w_{n,m} = \sum_{k \geq 1} \lambda_{n,k} (w_{n,m+k} + w_{n,m-k} - 2w_{n,m}).$$

By [theorems 6.1](#) and [6.2](#), the absolute value of the bracket is at most  $32 \cdot 4^{-n} k^2$ . Therefore

$$\begin{aligned} |\text{up}(\xi_{n,m}) - w_{n,m}| &\leq 32 \cdot 4^{-n} \sum_{k \geq 1} k^2 \lambda_{n,k} \\ &= 16 \cdot 4^{-n} \sum_{k \in \mathbb{Z}} k^2 \lambda_{n,k} \\ &= 16 \cdot 4^{-n} \left( \frac{n}{12} + \frac{1}{9} \right), \end{aligned}$$

which is (49).  $\square$

The proof separates two independent facts. The coefficient row has curvature  $\mathcal{O}(4^{-n})$  per lattice step, while the omitted Fourier tail averages over a discrete variance of order  $n$ . Their product is  $\mathcal{O}(n4^{-n})$ .

*Remark 7.2* (An elementary but weaker constant). If one avoids the Poisson calculation in [theorem 5.1](#), the support of  $q_n$  gives  $\lambda_{n,k} = 0$  for  $|k| > (n+2)/2$ . The same argument then yields

$$|\mathcal{W}_n(\xi_{n,m}) - \text{up}(\xi_{n,m})| \leq 4(n+2)^2 4^{-n}.$$

This already suffices for the sharp  $n2^{-n}$  law. The cardinal moment is what improves the intrinsic estimate from  $\mathcal{O}(n^2 4^{-n})$  to  $\mathcal{O}(n4^{-n})$ .

## 8 The sharp expansion for the Fabius sample

We now combine the exact drift with the center estimate. Recall  $\mathcal{A}_n$  from (28) and  $\delta_n$  from (29).

**Theorem 8.1** (Uniform second-order expansion). *For every  $n \geq 3$  and every  $x \in [0, 1]$ ,*

$$\boxed{\mathcal{A}_n(x) = F(x) + F'(x)\delta_n(x) + \frac{1}{2}F''(x)\delta_n(x)^2 + \rho_n(x)}, \quad (50)$$

where

$$|\rho_n(x)| \leq \left(\frac{4n}{3} + \frac{16}{9}\right)4^{-n} + \frac{4}{3}(n+2)^38^{-n}. \quad (51)$$

*Proof.* Put  $m = \lfloor 2^n x \rfloor$  and  $y = x - 1$ . Since  $m < 2^{n+1}$ , (25) and theorem 4.1 give

$$\mathcal{A}_n(x) = \mathcal{W}_n(\xi_{n,m}), \quad \xi_{n,m} = y + \delta_n(x).$$

By theorem 7.1, replacing  $\mathcal{W}_n$  by up at this center costs at most  $(4n/3 + 16/9)4^{-n}$ . Taylor's theorem and  $\|\text{up}'''\|_\infty \leq 64$  give

$$\text{up}(y + \delta) = \text{up}(y) + \text{up}'(y)\delta + \frac{1}{2}\text{up}''(y)\delta^2 + R, \quad |R| \leq \frac{64}{6}\delta^3.$$

By (31),  $\delta^3 \leq (n+2)^3/2^{3n+3}$ , so  $|R| \leq \frac{4}{3}(n+2)^38^{-n}$ . Finally  $\text{up}^{(r)}(x-1) = F^{(r)}(x)$  on  $[0, 1]$  by (1).  $\square$

For the leading rate it is convenient to absorb the quadratic Taylor term.

**Corollary 8.2** (Uniform first-order expansion). *For  $n \geq 3$  and  $x \in [0, 1]$ ,*

$$\mathcal{A}_n(x) = F(x) + F'(x)\delta_n(x) + r_n(x), \quad (52)$$

with

$$|r_n(x)| \leq \left(n^2 + \frac{16}{3}n + \frac{52}{9}\right)4^{-n} < (n+3)^24^{-n}. \quad (53)$$

*Proof.* Taylor's theorem to first order,  $\|F''\|_\infty \leq 8$ , and  $\delta_n(x) \leq (n+2)2^{-n-1}$  give a quadratic remainder at most  $(n+2)^24^{-n}$ . Add (49).  $\square$

Substituting (31) into (50) exposes three distinct scales.

**Corollary 8.3** (Binary-phase expansion). *Uniformly for  $x \in [0, 1]$ ,*

$$\begin{aligned} \mathcal{A}_n(x) - F(x) &= \frac{1}{2}F'(x)n2^{-n} + F'(x)(1 - \theta_n(x))2^{-n} \\ &\quad + \frac{1}{8}F''(x)n^24^{-n} + \mathcal{O}(n4^{-n}). \end{aligned} \quad (54)$$

*The implied constant is absolute.*

*Proof.* Expand  $\delta_n = (n/2 + 1 - \theta_n)2^{-n}$  and its square in (50). The cross term in  $\delta_n^2$  is  $\mathcal{O}(n4^{-n})$ , the constant term is  $\mathcal{O}(4^{-n})$ , and the remainder in (51) is uniformly  $\mathcal{O}(n4^{-n})$  because  $n^38^{-n} = o(n4^{-n})$ .  $\square$

The second term in (54) is not periodic in  $n$  for a general real  $x$ , but it is a literal binary shift:  $\theta_n(x)$  is the fractional tail left after moving the binary point  $n$  places. For a dyadic  $x$ , it vanishes eventually and the coefficient of  $2^{-n}$  becomes exactly  $F'(x)$ .

## 9 Pointwise and uniform sharpness

### 9.1 Uniform error profile and pointwise law

**Theorem 9.1** (Uniform normalized error profile). *For every  $n \geq 3$ ,*

$$\sup_{x \in [0,1]} \left| \frac{2^n}{n} (\mathcal{A}_n(x) - F(x)) - \frac{1}{2} F'(x) \right| \leq \frac{2}{n} + \left( n + \frac{16}{3} + \frac{52}{9n} \right) 2^{-n}. \quad (55)$$

Consequently

$$\frac{2^n}{n} (\mathcal{A}_n - F) \longrightarrow \frac{1}{2} F' \quad \text{uniformly on } [0, 1]. \quad (56)$$

In particular, for every fixed  $x \in (0, 1)$ ,

$$\lim_{n \rightarrow \infty} \frac{2^n}{n} (\mathcal{A}_n(x) - F(x)) = \frac{1}{2} F'(x). \quad (57)$$

Since  $F'(x) > 0$  on  $(0, 1)$ , the order  $n2^{-n}$  is sharp at every interior point.

*Proof.* From (31),

$$\frac{2^n}{n} \delta_n(x) = \frac{1}{2} + \frac{1 - \theta_n(x)}{n}.$$

Insert this in (52). Since  $0 \leq F' \leq 2$  and  $0 < 1 - \theta_n(x) \leq 1$ , the phase term is bounded by  $2/n$ . Multiplying (53) by  $2^n/n$  gives the second term on the right of (55).  $\square$

At the endpoints the limiting profile vanishes. Their unscaled errors obey a different, much faster regime; see [section 10](#).

### 9.2 The exact uniform constant

The upper bound follows immediately from  $0 \leq F' \leq 2$  and [theorem 8.2](#):

$$\|\mathcal{A}_n - F\|_{L^\infty[0,1]} \leq (n+2)2^{-n} + (n+3)^2 4^{-n}. \quad (58)$$

To prove that the coefficient is not an artifact of this estimate, we evaluate the midpoint exactly in [theorem 10.1](#). That formula gives

$$\|\mathcal{A}_n - F\|_\infty \geq (n+2)2^{-n} - 2^{-\binom{n}{2}}. \quad (59)$$

**Theorem 9.2** (Sharp uniform convergence rate). *For every  $n \geq 3$ , the corrected Thue–Morse prefix approximation satisfies the finite-level bracket*

$$-2^{-\binom{n}{2}} \leq \|\mathcal{A}_n - F\|_{L^\infty[0,1]} - (n+2)2^{-n} \leq (n+3)^2 4^{-n}. \quad (60)$$

Consequently

$$\|\mathcal{A}_n - F\|_{L^\infty[0,1]} = (n+2)2^{-n} + \mathcal{O}(n^2 4^{-n}) \sim n2^{-n}, \quad \lim_{n \rightarrow \infty} \frac{2^n}{n} \|\mathcal{A}_n - F\|_{L^\infty[0,1]} = 1. \quad (61)$$

*Proof.* Subtract  $(n+2)2^{-n}$  from (58) and (59). This gives (60); the asymptotic statements follow immediately. Alternatively, the last limit also follows by taking sup norms in the uniform profile limit (56), since  $\|F'/2\|_\infty = 1$ .  $\square$

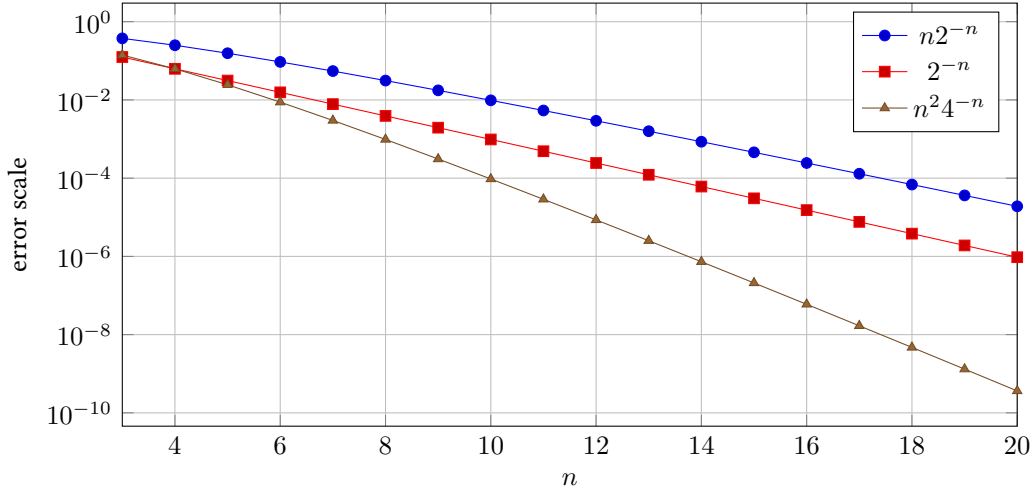


Figure 1: The three scales in (54). The deterministic center drift  $n2^{-n}$  dominates the one-cell scale  $2^{-n}$  and the quadratic Taylor scale  $n^2 4^{-n}$ .

## 10 Exact special values

The general estimates become especially transparent at 0,  $1/2$ , and 1.

### 10.1 The endpoints

Since  $a_{n,0} = 1$ ,

$$\mathcal{A}_n(0) = 2^{-\binom{n}{2}}. \quad (62)$$

At  $x = 1$ , the index is  $2^n$ . Using  $P_n = P_{n-1}(1 + z + \dots + z^{2^n-1})$  and  $g_{n-1} < 2^n$ , the coefficient at  $z^{2^n}$  is the sum of all coefficients of  $P_{n-1}$  except the constant term. Thus

$$\mathcal{A}_n(1) = 1 - 2^{-\binom{n}{2}}. \quad (63)$$

Hence

$$\mathcal{A}_n(0) - F(0) = 2^{-\binom{n}{2}}, \quad \mathcal{A}_n(1) - F(1) = -2^{-\binom{n}{2}}. \quad (64)$$

The endpoint convergence is superexponential in  $n$ , vastly faster than the interior  $n2^{-n}$  law.

### 10.2 An exact midpoint coefficient

**Proposition 10.1** (Exact midpoint error). *For every  $n \geq 2$ ,*

$$\boxed{\mathcal{A}_n\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{n+2}{2^n} - 2^{-\binom{n}{2}}.} \quad (65)$$

Consequently

$$\frac{2^n}{n} \left( \mathcal{A}_n\left(\frac{1}{2}\right) - \frac{1}{2} \right) = 1 + \frac{2}{n} - \frac{2^{n-\binom{n}{2}}}{n}. \quad (66)$$

*Proof.* Let  $b_r = [z^r]P_{n-1}(z)$  and  $M = 2^{n-1}$ . Since  $M < 2^n$ ,

$$[z^M]P_n(z) = \sum_{r=0}^M b_r.$$

Put  $D = g_{n-2} = 2^{n-1} - n$  and  $C = P_{n-2}(1) = 2^{\binom{n-1}{2}}$ . From  $P_{n-1} = P_{n-2}(1 + \dots + z^{2^{n-1}-1})$ , the central coefficients are

$$b_{D-1} = C - 1, \quad b_D = b_{D+1} = \dots = b_{2^{n-1}-1} = C, \quad b_{2^{n-1}} = C - 1.$$

The polynomial  $P_{n-1}$  is symmetric, has degree  $g_{n-1} = 2^n - n - 1$ , and total mass  $2^{\binom{n}{2}}$ . The interval  $[g_{n-1} - M, M] = [D - 1, M]$  is precisely the displayed central block. Pairing symmetric coefficients gives

$$\begin{aligned} 2 \sum_{r=0}^M b_r &= 2^{\binom{n}{2}} + \sum_{r=D-1}^M b_r \\ &= 2^{\binom{n}{2}} + (n+2)C - 2. \end{aligned}$$

Divide by  $2^{\binom{n}{2}}$  and use  $C/2^{\binom{n}{2}} = 2^{-(n-1)}$ . □

| $n$ | $\mathcal{A}_n(1/2) - 1/2$ | $(2^n/n)(\mathcal{A}_n(1/2) - 1/2)$ | leading value $1 + 2/n$ |
|-----|----------------------------|-------------------------------------|-------------------------|
| 4   | 0.3593750000               | 1.4375000000                        | 1.5000000000            |
| 6   | 0.1249694824               | 1.3330078125                        | 1.3333333333            |
| 8   | 0.0390624963               | 1.2499998808                        | 1.2500000000            |
| 10  | 0.0117187500               | 1.2000000000                        | 1.2000000000            |
| 12  | 0.0034179688               | 1.1666666667                        | 1.1666666667            |
| 14  | 0.0009765625               | 1.1428571429                        | 1.1428571429            |
| 16  | 0.0002746582               | 1.1250000000                        | 1.1250000000            |
| 20  | 0.0000209808               | 1.1000000000                        | 1.1000000000            |

The tiny discrepancy from  $1 + 2/n$  is exactly the superexponential last term in (66).

## 11 The corresponding result for Rvachev's bump

For  $0 \leq t < 1$ , let

$$\vartheta_n(t) = \{2^{n+1}t\}, \quad \Delta_n(t) = \frac{n+2-2\vartheta_n(t)}{2^{n+1}}. \quad (67)$$

Then

$$\xi_{n, \lfloor 2^{n+1}t \rfloor} = 2t - 1 + \Delta_n(t). \quad (68)$$

The proof of [theorem 8.1](#) gives, uniformly for  $0 \leq t < 1$ ,

$$\begin{aligned} \mathcal{J}_n(t) &= \text{up}(2t-1) + \text{up}'(2t-1)\Delta_n(t) \\ &\quad + \frac{1}{2} \text{up}''(2t-1)\Delta_n(t)^2 + \rho_n^{\text{up}}(t), \end{aligned} \quad (69)$$

with the same bound as (51). The right endpoint is exceptional because the coefficient identity stops one index earlier:

$$\mathcal{J}_n(1) = -2^{-\binom{n}{2}}, \quad \text{up}(1) = 0. \quad (70)$$

This exceptional error is negligible on every exponential scale discussed here.

**Corollary 11.1** (Pointwise and uniform rate for up). *For fixed  $t \in (0, 1)$ ,*

$$\frac{2^n}{n} (\mathcal{J}_n(t) - \text{up}(2t-1)) \longrightarrow \frac{1}{2} \text{up}'(2t-1). \quad (71)$$

At  $t = 1/2$ , both the derivative term and the actual error collapse:

$$\mathcal{J}_n(1/2) = 1 - 2^{-\binom{n}{2}}. \quad (72)$$

Moreover,

$$\lim_{n \rightarrow \infty} \frac{2^n}{n} \|\mathcal{J}_n - \text{up}(2 \cdot -1)\|_{L^\infty[0,1]} = 1. \quad (73)$$

The uniform lower bound may be read at  $t = 1/4$ , since  $\mathcal{J}_n(1/4) = \mathcal{A}_n(1/2)$  and  $\text{up}(-1/2) = F(1/2) = 1/2$ .

## 12 Rates for the histogram and for recentered interpolation

The sharp  $n2^{-n}$  rate belongs to the particular moving sample (28). It should not be attributed to the coefficient histogram itself.

### 12.1 The step histogram

Every interior point of a histogram cell lies within  $h_n/2$  of its center. Since  $\text{up}$  is 2-Lipschitz and the center error is given by [theorem 7.1](#),

$$\|\mathcal{W}_n - \text{up}\|_\infty \leq 2^{-n} + \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}. \quad (74)$$

The constant in front of  $2^{-n}$  is sharp. Near  $y = -1/2$ , where  $\text{up}'(y) = 2$ , move from a cell center to a point arbitrarily close to the right edge of the same cell. The histogram remains constant while  $\text{up}$  rises by  $h_n + o(h_n)$ ; the center error is  $o(h_n)$ . Therefore

$$\boxed{\lim_{n \rightarrow \infty} 2^n \|\mathcal{W}_n - \text{up}\|_\infty = 1.} \quad (75)$$

Thus the step representation has the expected one-cell rate  $2^{-n}$ .

### 12.2 Linear interpolation through the true centers

Extend  $w_{n,m}$  by zero for all  $m \in \mathbb{Z}$ , and let  $\mathcal{L}_n$  be the piecewise-linear function satisfying

$$\mathcal{L}_n(\xi_{n,m}) = w_{n,m} \quad (m \in \mathbb{Z}). \quad (76)$$

Let  $I_n \text{up}$  be the ordinary linear interpolant of the exact values  $\text{up}(\xi_{n,m})$  on the same mesh. Convexity of interpolation weights and [theorem 7.1](#) give

$$\|\mathcal{L}_n - I_n \text{up}\|_\infty \leq \left(\frac{4n}{3} + \frac{16}{9}\right) 4^{-n}.$$

For a  $C^2$  function on a mesh of width  $h$ , the linear interpolation error is at most  $h^2 \|f''\|_\infty / 8$ . Since  $\|\text{up}''\|_\infty \leq 8$ ,

$$\boxed{\|\mathcal{L}_n - \text{up}\|_\infty \leq \left(\frac{4n}{3} + \frac{25}{9}\right) 4^{-n}.} \quad (77)$$

Translating by one gives the same bound for an interpolant of  $F$  on  $[0, 1]$ .

This estimate removes both the step error and the deterministic  $n/2$ -cell drift. It is the most direct quantitative evidence that the iterated Thue–Morse coefficients are intrinsically accurate; the slower floor-sample rate is a coordinate choice.

| Approximation                 | Construction                                         | Uniform scale          |
|-------------------------------|------------------------------------------------------|------------------------|
| Center values                 | $w_{n,m}$ compared with $\text{up}(\xi_{n,m})$       | $\mathcal{O}(n4^{-n})$ |
| Linear center interpolant     | interpolate $(\xi_{n,m}, w_{n,m})$                   | $\mathcal{O}(n4^{-n})$ |
| Step histogram                | keep $w_{n,m}$ constant on its cell                  | $\sim 2^{-n}$          |
| Floor sample for $F$          | read $m = \lfloor 2^n x \rfloor$ without recentering | $\sim n2^{-n}$         |
| Centered finite spline        | different convolution/CDF scheme in the synthesis    | at most $2^{-n}$       |
| Literal unshifted prefix grid | use order $n$ rather than $n + 1$                    | fails at the endpoint  |

### 13 Why the constants take these values

The proof contains four constants, each with a transparent origin.

1. The coefficient curvature constant 32 is

$$4 \cdot 2^{\binom{n-2}{2} - \binom{n}{2}} = 32 \cdot 4^{-n}.$$

The factor 4 counts the four shifted copies of  $P_{n-2}$  in (47).

2. The discrete variance  $n/12 + 1/9$  consists of  $n$  unit-width box jitters, each of variance  $1/12$ , plus the variance  $1/9$  of the scaled Rvachev tail.
3. The leading displacement is  $n2^{-n-1}$ . Multiplication by the maximal slope  $F'(1/2) = 2$  gives the uniform leading error  $n2^{-n}$  and hence the sharp constant 1.
4. The quadratic Taylor coefficient uses the sharp bound  $\|F''\|_\infty \leq 8$ , while the cubic remainder uses  $\|F'''\|_\infty \leq 64$ . These are the  $r = 2$  and  $r = 3$  instances of (3), not fitted numerical estimates.

The hierarchy can be summarized schematically as

$$\underbrace{n2^{-n}}_{\text{degree/center drift}} \gg \underbrace{2^{-n}}_{\text{binary floor phase and one-cell error}} \gg \underbrace{n^2 4^{-n}}_{\text{quadratic Taylor term}} \gg \underbrace{n4^{-n}}_{\text{intrinsic center error}}. \quad (78)$$

For fixed  $x$ , the first two terms are explicitly visible in (54). The third is the first smooth nonlinear correction. The fourth is where the finite coefficient law itself differs from the infinite Rvachev law.

### 14 A Lean formalization roadmap

The new theorem is well aligned with the current module decomposition. The existing formalized chain is:

1. `ThueMorsePrefix.lean`: `iteratedPrefix`, the binomial convolution law, Prouhet cancellation, and exact dyadic zero runs;
2. `ThueMorseGenerating.lean`: formal power-series identities and the corrected normalized grid;
3. `ThueMorseApproximation.lean`: `iteratedPrefix_eq_approximationPolynomial_coeff`, `correctedPrefixCoefficient_eq_stepApproximant`, and the qualitative convergence theorems;
4. `StepApproximationLimit.lean`: coefficient monotonicity and the moving-argument squeeze used for pointwise convergence.

A formal proof of [theorem 9.2](#) can be organized into the following small additions.

1. Define the zero-extended normalized coefficient row  $w_{n,m}$  on  $\mathbb{Z}$  and prove (46) by coefficient extraction from (47). This is algebraic and should require no measure theory.
2. Reuse the existing cosine-product theorem to define  $q_n$  and prove  $\text{up} = \mu_n * q_n$ . The identity (35) is a direct reindexing plus the standard infinite cosine product.
3. Prove the cardinal partition  $\sum_k h q_n(kh) = 1$  by factoring one box density. The center averaging identity (41) then follows from convolution with an atomic measure.
4. For a first formal rate, use the compact support of  $q_n$  and obtain the elementary  $4(n+2)^2 4^{-n}$  center bound. This avoids differentiated Poisson summation and already proves the sharp  $n2^{-n}$  limit.
5. In a second refinement, invoke the repository’s Poisson-summation layer to prove the exact second moment (42); this upgrades the center estimate to (49).
6. Formalize the exact drift identity (30), apply the repository’s derivative bounds, and finish with a uniform Taylor estimate. The exact midpoint formula (65) supplies the matching lower bound without any compactness argument.

The logical core of the sharp constant requires only the weaker center estimate. Thus the project can be split cleanly: first formalize the asymptotic constant 1 with elementary support bounds, then strengthen the remainder by adding the cardinal second-moment calculation.

## 15 Conclusion

The iterated signed Thue–Morse prefixes approach the Fabius function through a sequence of exact algebraic identifications:

$$\text{prefix sums} \longleftrightarrow \text{coefficients of } P_n \longleftrightarrow \text{center heights of } \mathcal{W}_n \longrightarrow \text{up} \longrightarrow F.$$

The first two arrows are exact below the first omitted dyadic coefficient. The last two were already known qualitatively. The quantitative analysis shows that the finite coefficient law is exceptionally accurate at its natural centers: its error is only  $\mathcal{O}(n4^{-n})$ . The floor sample used to recover  $F$  reads those centers at a deterministic displacement

$$\delta_n(x) = \frac{n}{2^{n+1}} + \frac{1 - \{2^n x\}}{2^n}.$$

That displacement produces the complete leading behavior.

The final result is therefore both sharp and explanatory:

$$\mathcal{A}_n(x) - F(x) = F'(x)\delta_n(x) + \mathcal{O}(n2^2 4^{-n}), \quad \frac{2^n}{n}(\mathcal{A}_n - F) \longrightarrow \frac{1}{2}F' \quad \text{uniformly.}$$

Moreover

$$\|\mathcal{A}_n - F\|_\infty = (n+2)2^{-n} + \mathcal{O}(n^2 4^{-n}).$$

The leading pointwise profile is  $F'/2$ , the leading uniform constant is 1, and the next term remembers the binary tail  $\{2^n x\}$ . Recenter and interpolate the same coefficients, and the factor  $n2^{-n}$  disappears, leaving the much faster explicit bound  $\mathcal{O}(n4^{-n})$ .



## A Notation crosswalk

| Symbol             | Meaning                                                                         |
|--------------------|---------------------------------------------------------------------------------|
| $\varepsilon_m$    | signed Thue–Morse value $(-1)^{\text{wt}_2(m)}$                                 |
| $\sigma_k(m)$      | $k$ -fold inclusive prefix sum of $\varepsilon_m$                               |
| $P_n(z)$           | $\prod_{j=1}^n (1 + z + \cdots + z^{2^j-1})$                                    |
| $a_{n,m}$          | coefficient $[z^m]P_n(z)$                                                       |
| $g_n$              | degree $2^{n+1} - n - 2$ of $P_n$                                               |
| $h_n$              | mesh width $2^{-n}$                                                             |
| $\xi_{n,m}$        | centered coefficient location $(2m - g_n)/2^{n+1}$                              |
| $w_{n,m}$          | normalized height $a_{n,m}/2^{\binom{n}{2}}$                                    |
| $\mu_n$            | centered atomic coefficient distribution                                        |
| $\mathcal{W}_n$    | box-smoothed coefficient histogram $\mu_n * \beta_{h_n}$                        |
| $\mathcal{J}_n(t)$ | corrected bump sample $\sigma_{n+1}(\lfloor 2^{n+1}t \rfloor)/2^{\binom{n}{2}}$ |
| $\mathcal{A}_n(x)$ | Fabius sample $\mathcal{J}_n(x/2)$                                              |
| $\delta_n(x)$      | exact center displacement in (29)                                               |
| $\theta_n(x)$      | binary phase $\{2^n x\}$                                                        |
| $\lambda_{n,k}$    | cardinal tail weights $h_n q_n(kh_n)$                                           |

## B A compact statement of the main estimates

For rapid reference, for every  $n \geq 3$ :

$$\begin{aligned}
\sup_{m \in \mathbb{Z}} |\mathcal{W}_n(\xi_{n,m}) - \text{up}(\xi_{n,m})| &\leq \left( \frac{4n}{3} + \frac{16}{9} \right) 4^{-n}, \\
\|\mathcal{W}_n - \text{up}\|_\infty &\leq 2^{-n} + \left( \frac{4n}{3} + \frac{16}{9} \right) 4^{-n}, \\
\|\mathcal{L}_n - \text{up}\|_\infty &\leq \left( \frac{4n}{3} + \frac{25}{9} \right) 4^{-n}, \\
\sup_{x \in [0,1]} |\mathcal{A}_n(x) - F(x) - F'(x)\delta_n(x)| &\leq \left( n^2 + \frac{16}{3}n + \frac{52}{9} \right) 4^{-n}, \\
\sup_{x \in [0,1]} |\mathcal{A}_n(x) - F(x) - F'(x)\delta_n(x) - \tfrac{1}{2}F''(x)\delta_n(x)^2| &\leq \left( \frac{4n}{3} + \frac{16}{9} \right) 4^{-n} + \frac{4}{3}(n+2)^3 8^{-n}.
\end{aligned}$$

The normalized profile and norm satisfy

$$\begin{aligned}
\sup_{x \in [0,1]} \left| \frac{2^n}{n} (\mathcal{A}_n(x) - F(x)) - \frac{1}{2}F'(x) \right| &\leq \frac{2}{n} + \left( n + \frac{16}{3} + \frac{52}{9n} \right) 2^{-n}, \\
-2^{-\binom{n}{2}} &\leq \|\mathcal{A}_n - F\|_\infty - (n+2)2^{-n} \leq (n+3)^2 4^{-n}.
\end{aligned}$$

The exact asymptotic conclusions are

$$\begin{aligned}
\frac{2^n}{n} (\mathcal{A}_n - F) &\longrightarrow \frac{1}{2}F' && \text{uniformly on } [0, 1], \\
\frac{2^n}{n} \|\mathcal{A}_n - F\|_\infty &\longrightarrow 1, \\
2^n \|\mathcal{W}_n - \text{up}\|_\infty &\longrightarrow 1.
\end{aligned}$$

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